AENL 226 Power Electronics and power systems (Aug-Dec 2026)

Problem sheet 1

Problem 1

A 48 V to 12 V synchronous buck converter delivers 20 A at a switching frequency of 100 kHz. The high-side MOSFET has an on-resistance of 8 mΩ at operating temperature. Assume its drain current is approximately 20 A while it conducts. Its total turn-on plus turn-off energy is 120 μJ per cycle. Neglect gate-drive and output-capacitance losses. Find: (a) duty ratio, (b) conduction loss, (c) switching loss, and (d) total high-side MOSFET loss.

Solution:
Duty ratio: $D = V_o/V_{in} = 12/48 = 0.25$.
Conduction loss: $P_{cond} = I^2R_{DS(on)}D = 20^2 \times 0.008 \times 0.25 = 0.80$ W.
Switching loss: $P_{sw} = E_{sw}f_s = 120 \times 10^{-6} \times 100 \times 10^3 = 12.0$ W.
Total MOSFET loss: $P_{total} = 0.80 + 12.0 = 12.8$ W.

Problem 2

An IGBT in a single-phase inverter dissipates 38 W. The maximum permitted junction temperature is 150 °C, the maximum ambient temperature is 45 °C, the junction-to-case thermal resistance is 0.35 °C/W, and the case-to-sink thermal resistance is 0.15 °C/W. Assume that we want to keep a 20% thermal-resistance design margin. Thus, the selected heat sink will designed by limiting the selected heat sink to 80% of the calculated maximum sink-to-ambient resistance.
Find: (a) the maximum allowable sink-to-ambient thermal resistance, (b) the design target for heat-sink selection, and (c) the predicted junction temperature if a 1.6 °C/W heat sink is used.

Solution:
Allowable total junction-to-ambient resistance: $R_{\theta JA,max} = (T_{J,max} - T_A)/P = (150 - 45)/38 = 2.763$ °C/W.
Maximum sink-to-ambient resistance: $R_{\theta SA,max} = 2.763 - 0.35 - 0.15 = 2.263$ °C/W.
Design target with margin: $R_{\theta SA,target} = 0.80 \times 2.263 = 1.811$ °C/W. Therefore, select a heat sink rated at 1.81 °C/W or lower under the intended airflow and mounting conditions.
If we use a heat sink with $R_{\theta SA} = 1.6$ °C/W, total resistance is 0.35 + 0.15 + 1.60 = 2.10 °C/W. Predicted junction temperature: $T_J = 45 + 38 \times 2.10 = 124.8$ °C.

Problem 3

A boost converter raises 200 V to 400 V and supplies 2 kW. Assume ideal power balance for estimating current and continuous conduction. The switching frequency is 50 kHz.

The MOSFET has $R_{DS(on)} = 0.18\ \Omega$ and total switching energy $E_{sw} = 0.70$ mJ per cycle. The output diode has a constant forward drop of 1.5 V.

Find: MOSFET conduction loss, MOSFET switching loss, diode conduction loss, total semiconductor loss.

Solution:
Duty ratio: $D = 1 - V_{in}/V_o = 1 - 200/400 = 0.50$.
Input/inductor current average: $I_L \approx P_o/V_{in} = 2000/200 = 10$ A.
MOSFET conduction loss: $P_{M,cond} = I_L^2 R_{DS(on)} D = 10^2 \times 0.18 \times 0.50 = 9.0$ W.
MOSFET switching loss: $P_{M,sw} = 0.70 \times 10^{-3} \times 50 \times 10^3 = 35.0$ W.
Diode conduction loss: $P_{D,cond} = V_F I_L (1 - D) = 1.5 \times 10 \times 0.50 = 7.5$ W.
Total semiconductor loss = 9.0 + 35.0 + 7.5 = 51.5 W.

Problem 4
Four identical MOSFET modules are mounted on one heat sink. Each module dissipates 22 W and has $R_{\theta JC} = 0.45$ °C/W and $R_{\theta CS} = 0.10$ °C/W. The maximum ambient temperature is 40 °C, and the design junction-temperature limit is 125 °C. Assume the heat sink is isothermal and all four devices transfer their heat into it.
Find: (a) the maximum allowable sink-to-ambient thermal resistance and (b) the resulting heat-sink and junction temperatures if a 0.65 °C/W heat sink is selected.

Solution:
Total heat into the sink: $P_{sink} = 4 \times 22 = 88$ W.
For one module, the case-path temperature rise above the sink is $\Delta T_{JS} = 22(0.45 + 0.10) = 12.1$ °C.
Thus the maximum permissible sink temperature is $T_{S,max} = 125 - 12.1 = 112.9$ °C.
Maximum sink-to-ambient resistance: $R_{\theta SA,max} = (112.9 - 40)/88 = 0.828$ °C/W.
With a 0.65 °C/W heat sink: $T_S = 40 + 88 \times 0.65 = 97.2$ °C.
Junction temperature of each module: $T_J = 97.2 + 22(0.45 + 0.10) = 109.3$ °C.

Problem 5
A power MOSFET in a 5 kW converter carries an RMS current of 25 A. Its effective $R_{DS(on)}$ at operating temperature is 24 mΩ, and its conduction duty cycle is 0.60. The combined turn-on and turn-off energy is 0.50 mJ per switching cycle. The MOSFET has $R_{\theta JC} = 0.30$ °C/W and $R_{\theta CS} = 0.12$ °C/W. Ambient temperature is 50 °C and the junction-temperature limit is 140 °C.

Find: At 20 kHz and 60 kHz, calculate (a) MOSFET conduction loss, (b) MOSFET switching loss, (c) total converter-associated loss from these terms, and (d) the

maximum allowable heat-sink thermal resistance for the MOSFET. State which frequency requires the more demanding thermal design.

Solution:
Conduction loss at either frequency: $P_{cond} = I_{RMS}^2 R_{DS(on)} D = 25^2 \times 0.024 \times 0.60 = 9.0$ W.

At 20 kHz:
$P_{sw} = 0.50 \times 10^{-3} \times 20 \times 10^3 = 10.0$ W.
Total associated loss = 9.0 + 10.0 = 19 W.
$R_{\theta SA,max} = (140 - 50)/19.0 - 0.30 - 0.12 = 4.32$ °C/W.

At 60 kHz:
$P_{sw} = 0.50 \times 10^{-3} \times 60 \times 10^3 = 30.0$ W.
Total associated loss = 9.0 + 30.0 + = 39 W.

$R_{\theta SA,max} = (140 - 50)/39.0 - 0.30 - 0.12 = 1.89$ °C/W.

At 20 kHz, the associated loss is 19.054 W and the heat sink must be 4.32 °C/W or lower. At 60 kHz, the associated loss is 39.162 W and the heat sink must be 1.89 °C/W or lower. The 60 kHz design is much more demanding thermally because switching loss triples while conduction loss remains unchanged.

Problem 6

The voltage across and current through a load are shown in Fig. 1(a) and (b). Both the waveforms are periodic with T=20 ms.

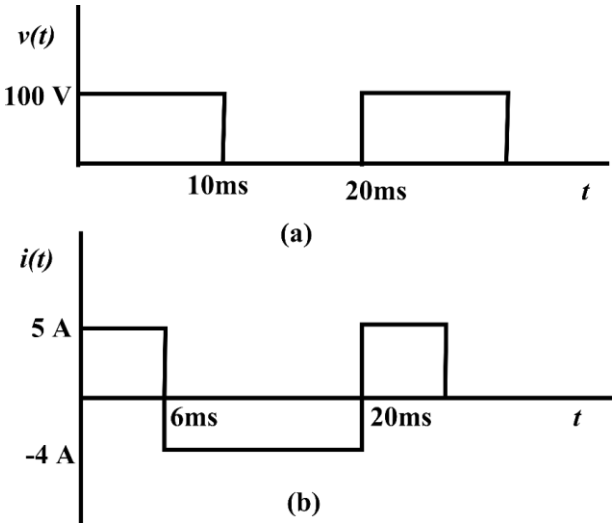


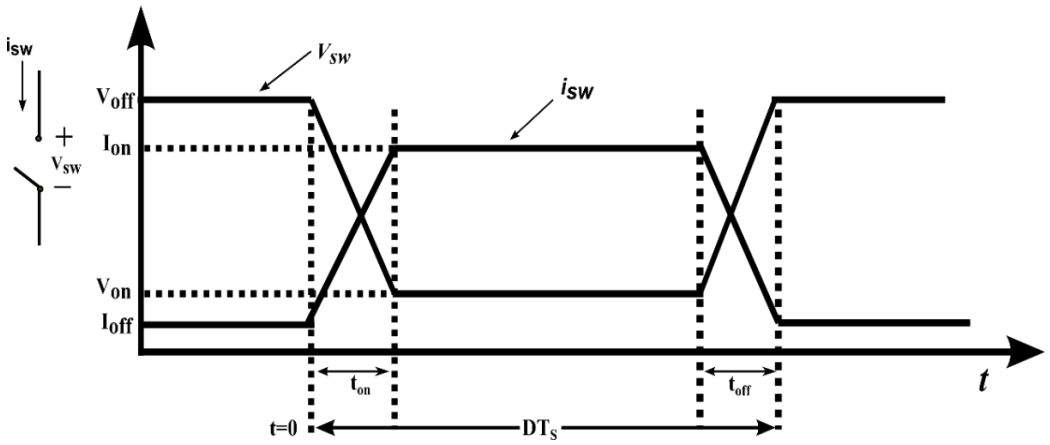
Fig.1

Determine (a) the instantaneous power, (b) the average power, and (c) the energy absorbed by the load in each period.

Ans (a) 500W,-400W, 0W (b) 70W (c) 1.4J

Problem 7

Consider a power switching device whose voltage and current waveforms are shown in Fig. 2. Assume V_{off} =100V, V_{on} = 0.7V, I_{on}=5 A, I_{off} =0.2 A, D =1, T_s =150μs, t_{on}= 5μs and t_{off} =10 μs.

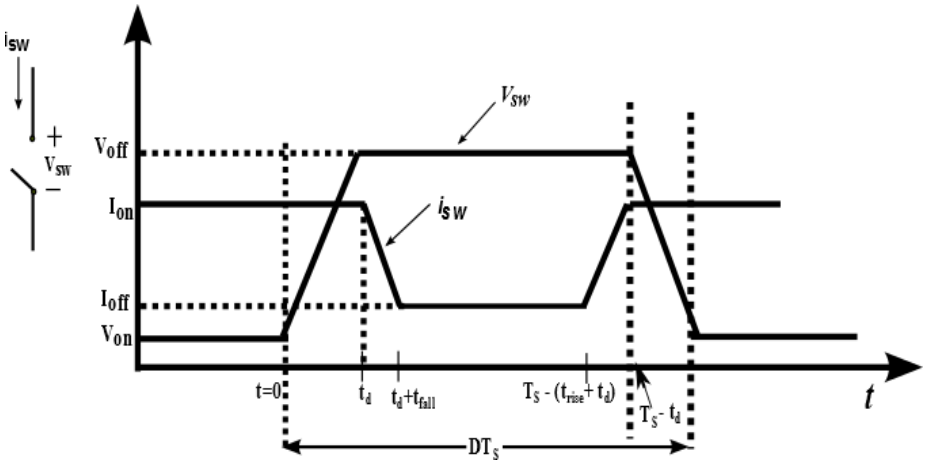


(a) Find an expression for the instantaneous switching and conduction power losses in the device.
(b) Determine the energy dissipated in the device during one switching period.
(b)Determine the average power dissipated in the device during one switching period.

Ans: (b) 1722uJ (c) 11.5 W

Problem 8

Consider a power switching device whose voltage and current waveforms are shown in Fig.3. Assume t_{rise}=140ns, t_{fall}=200ns, t_d=80ns, D=1, f_s=10 kHz, V_{on} = 0V, I_{off} =0 A V_{off} =100 V and I_{on}=10A.
(a) Find an expression for the instantaneous switching and conduction power losses in the device.
(b) Determine the average power dissipated in the device during one switching period.



Ans: (b) 25W

Problem 9

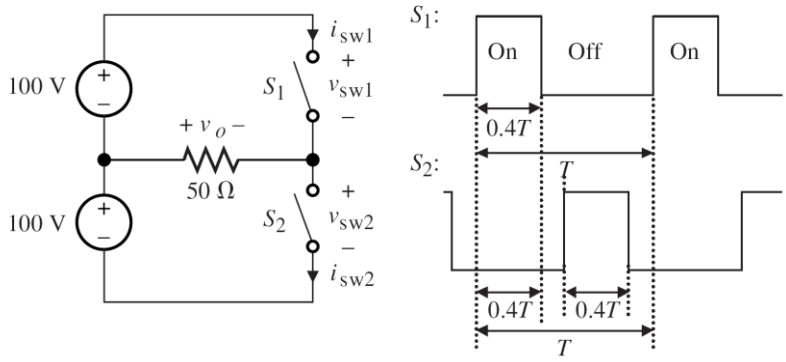
Consider the two switch circuit shown in Figure, where S1 and S2 are assumed ideal.

(a) Sketch isw1, vsw1, isw2, vsw2.

(b) Find out the average output voltage.

(c) Find out the average power delivered to the load.

(d) Find out the voltage and current rating of switch S1 and S2.



Ans: (b) 0 V (c) 160W

Problem 10

The power dissipated in a device is 20 W. The thermal resistance of the device is 1^0C/W. Assume the thermal interface material has thermal resistance of 1.5^0C/W. To maintain the junction temperature below 100^0C for 25^0C ambient temp, what should be the thermal resistance of the heat sink?

Ans: 1.25 ^{0}C /W

Solve few more problems from Issa Batarseh book (textbook).